

Needfinding

HCI Fundamentals

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Hall of Fame or Shame?

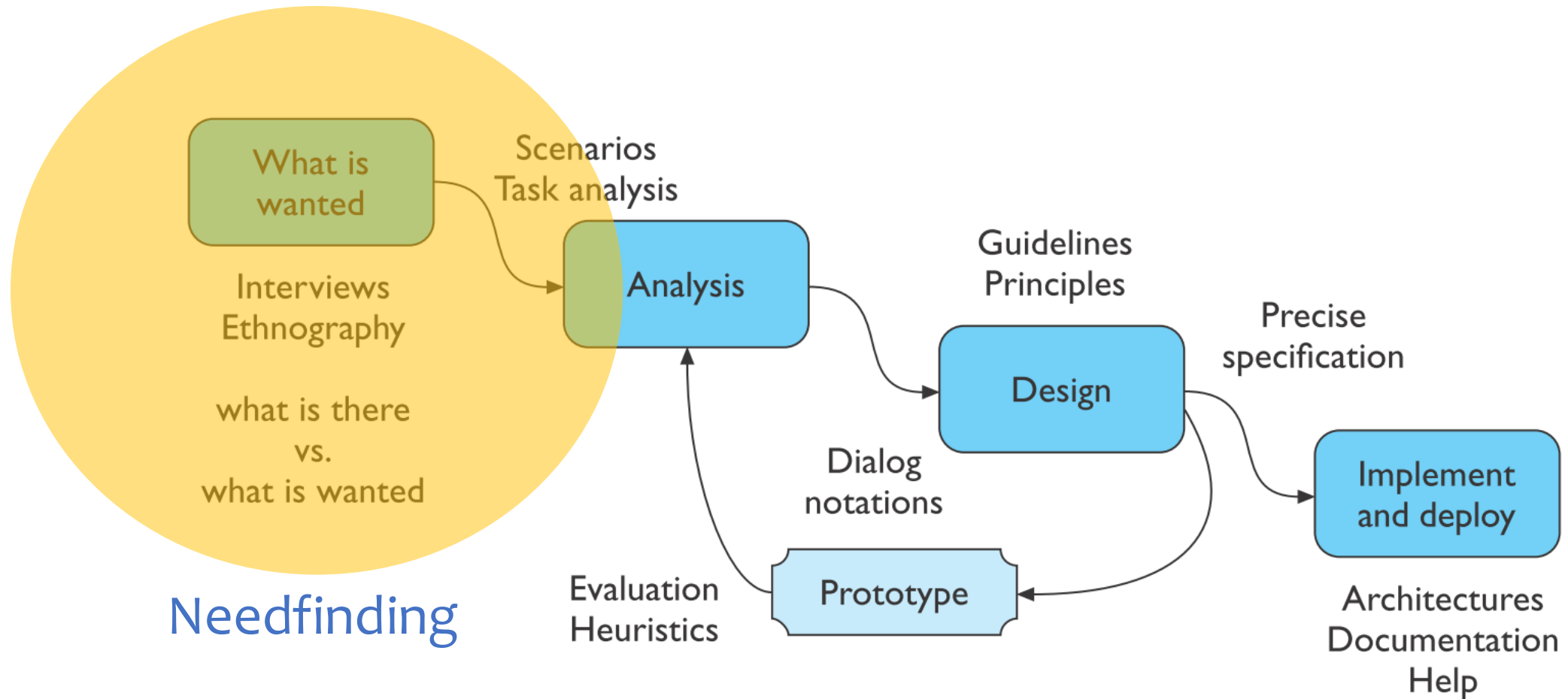


<https://www.youtube.com/watch?v=e2RoNSKtVAo>

Goals

- Finding and framing problems, starting from people
- Understanding user needs (and system requirements)
 - “Needfinding”
- Knowing tools and instruments for needfinding

Human-Centered Design Process



What Are Needs?

- “Human emotional or physical necessities”
- “Gaps in a system”
- They are **verbs**: activities and desires with which your user could use help
 - not nouns (solutions)
- Often, it is helpful to use the phrases ‘needs a way to’ or ‘needs to be able to’ to describe user needs
- They emerge directly from user traits or from contradictions between two traits
 - such as a disconnect between what they says and what they do

What is Needfinding?

- Needs: **gaps** in a system
- Needfinding: discovering opportunities by **recognizing** this gaps



What is Needfinding?

- Figure out the story of **what** and **why**...
- ... and tell a new one!



Main Needfinding Questions

- Needfinding = **Finding Potential User Needs**
 - What do users need?
- That also requires
 - Who are the users?
 - How are they doing it, now?
 - What is the context in which they are doing it?
 - Can't we just ask them?

Know Your Users (1)



- Who are the users of the system?
 - Uniform, or different categories/groups?
 - Young/old? Novice/experienced?
 - Do not think of “generic” users, split the categories
- ***You*** are not a [representative] user
 - Designers and developers’ skills, knowledge, attitude, background, interests, ... are totally unlike those of your users
 - Except by chance (e.g., you are also students, developers, ...)
- The **client** is not a [representative] user
 - Bosses, managers, directors, ... believe they know their employees and their jobs. Actually, they don’t
 - Always seek the actual users that will use the system

Know Your Users (2)

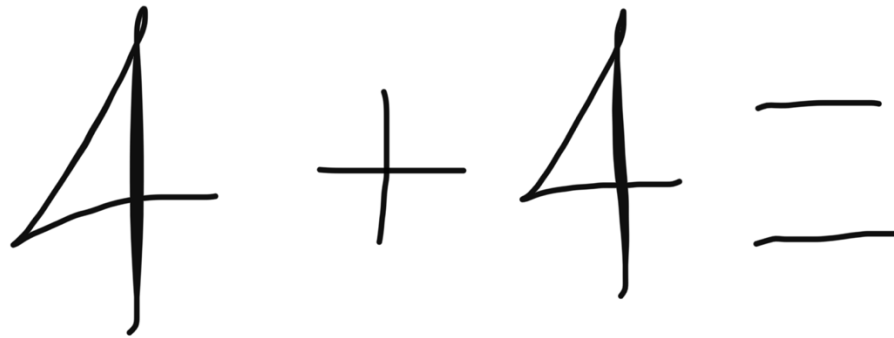
- Talking to users
 - Surveys
 - Interviews
 - Direct involvement (participatory design)
 - Bypass corporate policies
 - Understand real current behavior, pain points, workarounds, ...
- Watching users
 - Observation sessions
 - Video recording (and analysis)
 - Diaries
 - Analyze their work (artifact, processes, action sequences)
 - Discuss with users the findings of the observation (may discover the “why”)

Know Your Users (3)

- Imagining users
 - When real users are not available
 - Imagine how a real user would behave (very difficult)
 - Building “imaginary” users: *personas*
 - Detailed description of hypothetical persons in a given role
 - Imagine them as they were a real person

Problem Framing

- Framing problems is as important as finding them
 - we need to check our and other people's **assumptions** to fully understand the needs, the context, and emphasize with them
- For example, let's say that a person shows this to us:



A handwritten mathematical expression consisting of the number 4, followed by a plus sign, another 4, and an equals sign. The numbers are drawn with a single vertical stroke and a diagonal stroke, while the plus and equals signs are simple horizontal and vertical lines.

$$4 + 4 =$$

Needfinding Methods

Techniques adopted in Needfinding and Requirement Analysis

Many Methods!

- Observations
- Diaries
- Interviews
- Focus groups
- Surveys
- Contextual Inquiry

To Get Started...

After having identified and found the users to *<insert needfinding method>*:

1. Introduce yourself and try to make the participant comfortable.
2. Discuss confidentiality and get *explicit consent* for any filming or recording you may be doing.
3. Begin to introduce the topic/goal you are interested in understanding.
4. Start *<insert here needfinding method>*.

Observation

“You can observe a lot, just by watching” – Yogi Berra

Ethnographic Observation

- Embed in the users' environment, culture, behavior
- Goal: to obtain the necessary data to influence interface (re)design
- Learn the language of users and their environment
- Listen and observe carefully
 - Sometimes ask questions and clarifications
- Audio-Video Record / Take Notes
- Risks:
 - Misinterpret observations
 - Disrupting normal practice
 - Overlooking important information

What Should Learn By Observation? (1)

1. What do people do now?
2. What values and goals do people have?
3. How are these activities embedded in a larger ecology?
4. Similarities and differences across people
5. Other types of context, like time of day

Especially tacit (unspoken) knowledge

What Should Learn By Observation? (2)

- Process vs. Practice
- Process: how things are officially supposed to happen, and are officially part of the training
- Practice: set of workarounds, practical tricks, information learnt from the field and from experience, etc., that are part of daily activities



Types of Observation (1)

Controlled Observation, within a Lab environment

- **Easy to reproduce.** If you use a quantitative approach – easy to get similar results by repeating
- **Easy to analyze.** Quantitative data requires less effort to analyze than qualitative data.
- **Quick to conduct.** Recruitment may take a little time, the controlled observation is fairly fast to run.
- **The Hawthorne Effect.** The act of observation of how someone does something can change their approach to carrying out the task.

Types of Observation (2)

Naturalistic Observation: studying the user “in the wild”, less structured

- **More reliable.** When people use a product in real life – they are much more likely to encounter the frustrations (and benefits) of real life use than they are in a lab following a set of instructions.
- **More useful for ideation.** Qualitative research can generate lots of ideas for product improvement as it opens up possibilities that aren’t found in quantitative research.
- **Difficult to include a representative sample.** More expensive and time consuming to conduct, limits the reach of the research. Use this kind of research to create ideas and then test those ideas with other forms of research.
- **Difficult to make them replicable.** Problem with sample sizes and dependence on the observer.
- **Hard to manipulate external variables.** For example, if it’s raining when you observe your users working on a smartphone – their behavior is likely to be different to when it’s sunny. You have no control over the weather “in the wild”.

Blending In

- **Becoming part of the wall**
(Complete Observer)
 - Avoid being intrusive or modifying behaviors
 - Avoid video-recording or interruptions
 - Schedule time for discussing your observations
- **Becoming “one of them”, like a spy**
(Complete Participant)
 - Undergo training process
 - Official information
 - Matter-of-fact information shared by co-workers
 - Observe all the practices
 - Validate your observations with the others

Data Collection

Subjective

- Impressions
- Ranking/ratings by users on different questions
- Written summary report
- Artifacts and “hints” in the workplace

Objective

- Anecdotes
- Critical incidents
- Observed errors
- Observed workarounds



Interviewing

Asking users about their needs and desires...

... what could possibly go wrong?

Main Forms of Interviews

- In-person interviews
 - Time-demanding, in-depth knowledge
 - Structured vs Unstructured
 - One-to-one vs. Focus groups
- Surveys
 - Fast, more superficial
 - Sets of questions with predefined possible answers
 - Paper-based or on-line

Beware!

- Users do not know what they want
 - Maybe subconsciously, but not rationally
 - They will tell what they think you like to hear
 - Especially for “new” products or “disruptive” technologies
 - They lack the creativity or the technical expertise to understand the new product
 - They take the current context for granted (e.g., required workarounds “because, yes”)



Stop Asking Users What They Want

<https://uxplanet.org/stop-asking-users-what-they-want-21e9ba646bce>

Choosing Participants For Interviews

- Representative of target users
 - [All] Stakeholders
- May be current users of a similar system
- Might also be the non-users (for a new product)
- Approximate, if necessary (with similar users)
- Incentives, motivations, small gifts

Executing Interviews

- Schedule a time and place comfortable for users
- Introduce yourself, explain your purpose
 - You are not testing them; they are helping you
- Begin with open-ended, unbiased, non-leading questions
- Ask the question and let them answer
 - Give enough time. The 2nd reply is often more interesting than the 1st.
- Follow-up with related questions. Deep dive into interesting points



Interview

<http://dlrtoolkit.com/interview/>

Guidelines For Questions

- Structured questions are easier to process, unstructured questions solicit more comments
- Open-ended questions, with follow-up discussion
- For quantitative questions (e.g., rate in a scale 1 to 5, such as Likert), always ask what they mean by “4”
- Aim at direct, concrete, specific questions that ask for detailed answers
- Use the language of the user
- Always try the question with a smaller (trusted) group, for debugging

Examples of Open-Ended Questions

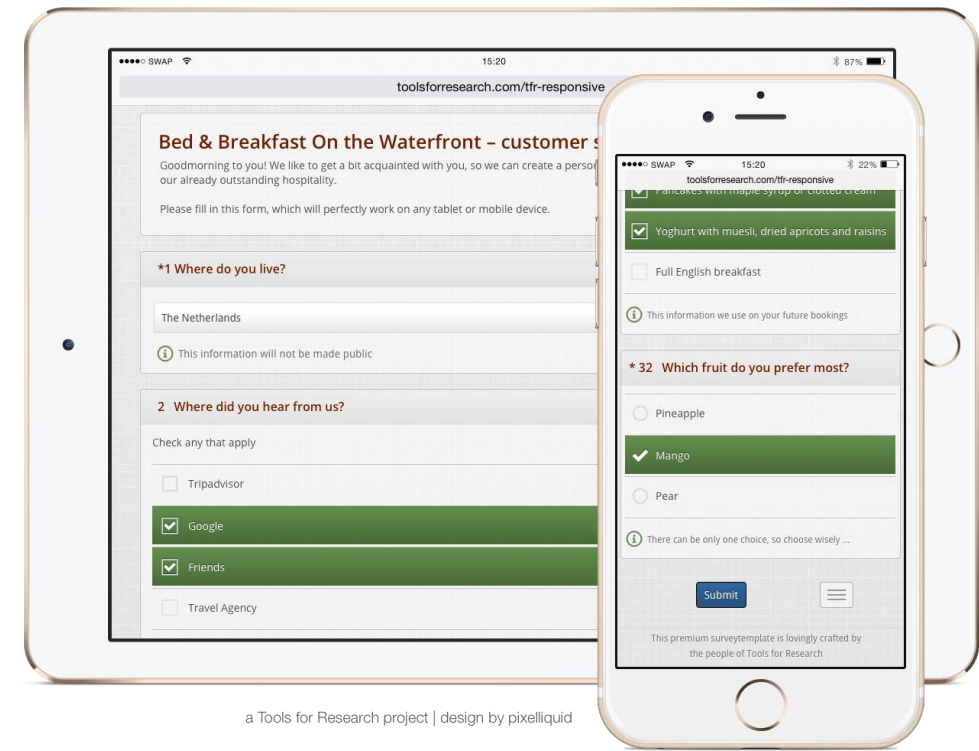
- ‘Tell me about your typical day.’
- Tell me three good things about. ...’
- ‘and three bad things.’
- ‘What has gone wrong with the application recently? How did you cope?’
- ‘What else should we have asked about?’

Bad questions – To Avoid

- Why is feature [x] important to you?
 - ‘Leading’ question
- What would you like in a tool?
 - User are experts in their domain, not expert in design
- What do you like in [x]?
 - Assuming question. Maybe he doesn’t like it
- What would you do in a hypothetical situation?
 - Users cannot imagine the complete environment or an unusual situation

Bad questions – To Avoid

- How often do you do [x]?
 - Humans are very bad at estimating (and biased in the answers)
 - May obtain by log analysis (if an application already exists)
- Binary questions (yes/no)
 - Do not yield motivation
- ‘Tell me a story about you’
- ‘How do you reach the decision? Did you meet? Did someone decide without you? ...’
 - Avoid suggesting possible answers. Trust the question: ask it and wait in silence



a Tools for Research project | design by pixelliquid

Surveys

Requirements gathering through (on-line) surveys and questionnaires

On-line surveys

- Familiar, cost-effective, potentially reaching a very wide audience
- Results can be easily visualized and analyzed with statistical methods
- Should be very careful with setting the goal of the survey
 - First define what statistics and/or charts you need, then design the survey structure and questions

Risks in surveys

- Good for a shallow view over a large base of respondent, but not for a “deep” analysis
- Impossible to ask follow-up questions
- Biased data if questions rely on user’s memory or on “sensitive” issues (money, emotions, ...)
- Finding a representative target population
 - Stratification

Survey structure

- Declare the purpose of the survey and the expected time
- One or more sections with targeted questions
- Background information about the users
 - Limit “mandatory” fields to a minimum

Background information (examples)

- Background demographics (age, gender, origins, native language, education, income)
- Experience with computers (specific applications or software packages, length of time, depth of knowledge, whether knowledge was acquired through formal training or self-teaching)
- Job responsibilities (decision-making influence, managerial roles, motivation)
- Personality style (introvert versus extrovert, risk taking versus risk averse, early versus late adopter, systematic versus opportunistic)
- Reasons for not using an interface (inadequate services, too complex, too slow, afraid)
- Familiarity with features

Types of Questions

- Open-ended questions
 - Solicit specific motivation, to avoid too generic answers
 - Require methodology for analyzing the responses
- Closed-ended questions
 - Only one possible choice
 - **Ordinal** values: a scale of ordered possibilities (e.g., from 1 to 5)
 - **Nominal** values: alternatives, with no ordering relationships (e.g., city or department)

Measurement scales (1)

- Nominal scale
 - Predefined set of distinct (pairwise disjoint) classes. No ordering.
 - Example: Gender. City of residence. Course of enrollment. Preferred color. Room of your house.
 - Meaningful statistics: Counting. Frequency. Mode.
- Ordinal scale
 - Distinct classes, pre-defined ordering. No defined “distance” across the different values
 - Example: University scores. Preference (Likert). Stars. Thumbs up/down.
 - Meaningful statistics: Ordering. Median. Quartiles. Percentiles. Rank.

Measurement scales (2)

- Interval scale (rarely used)
 - Numerical value, arbitrary zero-value, arbitrary unit of measure
 - Example: Date. Hour. Temperature.
 - Meaningful statistics: Mean (Average). Variance.
- Ratio scale
 - Numerical value, fixed zero-value, arbitrary unit of measure
 - Example: Duration of a task. Length of an object. Age.
 - Meaningful statistics: All (higher-order, geometric mean, ...)
- Absolute scale
 - Numerical (integer) value, corresponding to the cardinality of a set
 - Example: Number of employees
 - Meaningful statistics: All

Ordinal scales: Likert scales

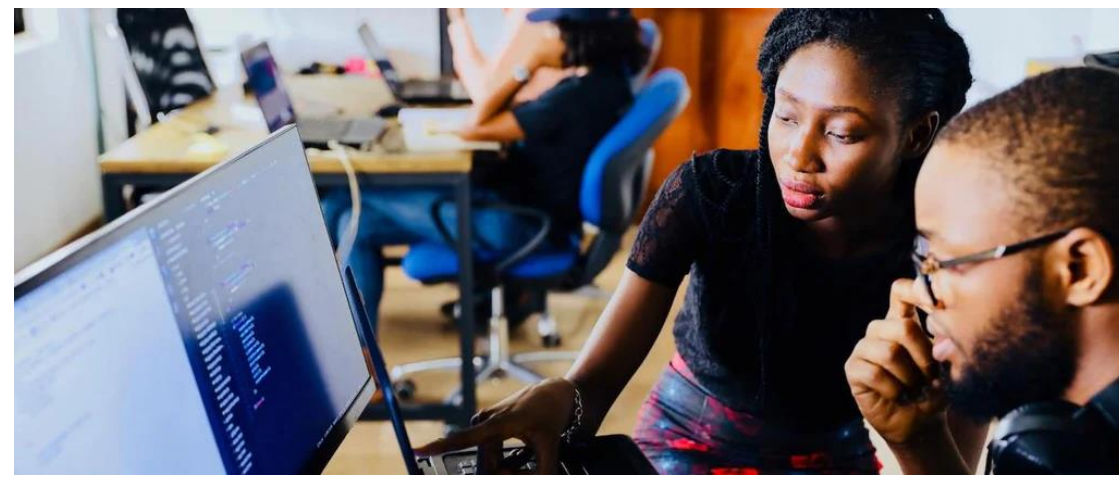
- Ask for the level of agreement about a statement
- Extreme values are very rare to be selected
- An even number of levels prevents a “neutral” response
- 4-level: Strongly Disagree / Disagree / Agree / Strongly agree
- 5-level: Strongly Disagree / Disagree / Neither agree nor disagree / Agree / Strongly agree
- Numerical ranges: 1-5, 1-7, 1-9



https://en.wikipedia.org/wiki/Likert_scale
<https://www.simplypsychology.org/likert-scale.html>

Warnings

- If possible, use two simpler questions instead of one complex one (separate the factors of interest)
- Avoid negative words in the question
- Biased questions solicit biased responses



Contextual Inquiry

Observing while interviewing

Contextual Inquiry

- Contextual inquiry = inquiry of context
 - *Context*: the research takes place in the users' natural environment as they conduct their activities the way they normally would
 - *Inquiry*: the researcher watches the user as she performs their task and asks for information to understand how and why users do what they do
- Participants are observed while they perform tasks **and** talk about what they are doing while they perform them
 - Observation mixed with interviews
- Participants take a more *active* role in leading their session, like a craftsman with their apprentice

Contextual Inquiry Vs. Interviews

- Interviews and surveys rely on the users' ability to recall and explain a process that they are removed from in that moment
 - People attempt to summarize their processes, but important details like reasoning, motivation, and underlying mental models are *left out* of this summary
 - Often you only obtain a superficial understanding of the users' approach to the activity
- However, people can easily talk about **what** they are doing and **why**, when they are doing it
 - You get to see the interruptions, superstitious behaviors, and illogical processes

Contextual Inquiry Vs. Observations

- Contextual inquiry works well for understanding **in-depth thought processes** of users
 - and the underlying structure of their activities
- Direct observation works better when:
 - The activity does not require in-depth thoughts, e.g., designing a standard e-commerce page
 - People cannot (or should not) be interrupted or distracted
 - Think at doctors in hospitals, for instance
 - In these cases, a valid option is an observation followed by an interview

Risks and Downsides (I)

- **Participants always go into “interview mode”**
 - Working and demonstrating while being interviewed is unfamiliar to most people
 - It can be easy for participants to begin summarizing their processes at the end
- **It is a show-and-tell of frustrations with the current solution**
 - Participants may feel you are looking for their feedback on all the problems with the system
 - However, the purpose of contextual inquiry is to understand people’s thoughts and work process, regardless of the solution

Risks and Downsides (II)

- **You introduce biases**

- It is important to approach contextual inquiry objectively, without any preconceived notions or opinions about the subject matter
- Go into the activity with an open mind, treating everything you learn with the same level of importance

- **You bias the user**

- As you perform the contextual inquiry, it is possible that the participant may adjust their process to fit into the discussion or your interpretations

References

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- Contextual Inquiry, <https://www.nngroup.com/articles/contextual-inquiry/>

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